

Database Management System

BCA — Semester IV — 2024 — Answer Sheet

Subject Code: CACS255

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group C Attempt any TWO questions[2x10=20]

2. Define database system. Write the merits and demerits of DBMS.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Draw and explain three schema architecture of the database system.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. What are an entity and its attributes? Write the difference between a strong and weak entity set with an example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Why integrity constraints are used in databases? Explain different types of integrity constraints with examples.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. What is functional dependency? Explain its types with examples.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Differentiate between discretionary access control and mandatory access control with example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Write the ACID properties of transactions. Explain the concepts of serializability in database transactions.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Why normalization is required in database? Explain the 1NF, 2NF and 3NF of normalization with example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. How participation constraints are used in ER diagram? Explain. Also Construct an E-R diagram for a car insurance company whose customers own one or more cars each. Each car has associated with it zero to any number of recorded accidents. Each insurance policy covers one or more cars and has one or more premium payments associated with it. Each payment is for a particular period and has an associated due date, and the date when the payment was received.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Consider the following employee database, where primary keys are underlined Supplier (supplier-id, supplier-name, city) Suppliers (Supplier-id, parts-id, quantity) Parts (part-id, par-name, color, weight)a) Write a SQL query to create above table.b) Find the name of all suppliers who has supplied parts with quantity more than100 and the name of all parts supplied by "RD Traders"c) Find the number of parts supplied by "S02"d) Find the name of suppliers located in city "KTM".

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.